Vrije Universiteit Amsterdam

Bachelor Thesis

Cardinality-Constrained Portfolio Optimization:

A Genetic Algorithm Out-of-Sample Evaluation

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Abstract

Context. Mean-variance optimization is theoretically appealing but disappoints out-of-sample when the asset universe is large relative to the estimation window. The sample covariance matrix becomes poorly conditioned, and the optimizer fits noise. Cardinality constraints may reduce this problem by limiting how many covariance terms need accurate estimation, but they make the problem NP-hard.

Goal. This thesis evaluates a genetic algorithm for cardinality-constrained portfolio optimization out-of-sample against classical mean-variance benchmarks and an equal-weight strategy.

Method. The Genetic Algorithm (GA) is applied to roughly 867 US stocks per month over 252 monthly periods (2005–2025) with a 60-month estimation window, cardinality $K \in [10, 30]$, and weights in $[0.02, 0.15]$. Transaction costs of 0.3% per unit of turnover are deducted from all returns. Hyperparameters are tuned with Optuna on an initial 2005–2012 window, then fixed. This overlap is treated as a methodological limitation. Three benchmarks are evaluated: unconstrained Mean-Variance Optimization (MVO), constrained MVO with a 0.15 weight cap, and equal-weight $1/N$. A fixed-K sensitivity analysis covers $K \in \{10, 15, 20, 25, 30\}$.

Results. Both MVO variants achieve near-identical net Sharpe ratios (0.5810 and 0.5809). The 0.15 cap rarely binds across 867 stocks. The adaptive GA achieves 0.2741, below all three benchmarks. A 252-period ablation ($\lambda = 0$ versus tuned $\lambda = 1.8437$) shows the penalty is over-aggressive: removing it raises the GA's net Sharpe to 0.4993 and average cardinality from 16.3 to 25.1 stocks. Fixed-K configurations at $K \in \{25, 30\}$ reach Sharpe ratios of 0.4400 and 0.4363.

Conclusions. The adaptive GA underperforms all three benchmarks. The primary cause is an over-aggressive turnover penalty: $\lambda = 1.8437$ suppresses cardinality to $\bar{K} = 16.3$ and costs more in foregone return than it saves, about three-quarters of the Sharpe gap against MVO. A secondary factor is in-sample fitness noise: even after Ledoit-Wolf shrinkage, a covariance estimated from 60 monthly observations for roughly 867 stocks ($N/T \approx 14.5$) remains a noisy basis for ranking sparse portfolios.

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1 Introduction

Portfolio management requires balancing expected return against risk, a problem Markowitz (1952) formalised as quadratic optimization over asset weights. Given accurate estimates of expected returns and covariances, mean-variance optimization should identify the portfolio that maximises risk-adjusted return. In practice those estimates come from historical data, and with large asset universes the estimation error can be severe. Michaud (1989) shows that standard MVO amplifies this noise rather than extracting signal, assigning the largest weights to the assets whose returns are least certain, so the resulting portfolio fits historical noise and performs poorly out-of-sample. DeMiguel et al. (2009) quantify the failure: across seven datasets, the naive equal-weight strategy matches or outperforms 14 optimization models out-of-sample, including shrinkage and factor-model variants.

One response is to constrain the optimization. Imposing weight upper bounds reduces a portfolio's sensitivity to noisy return estimates (Jagannathan and Ma, 2003). A stronger form is the cardinality constraint: restricting the number of assets held to some $K \in [K_{\min}, K_{\max}]$ limits how many covariance terms must be estimated accurately, curbing how much estimation error reaches the final portfolio. This comes at a computational cost. Cardinality constraints make the problem NP-hard: the choice of which K assets to hold and how to weight them cannot be solved jointly by standard convex optimizers, so heuristic methods are needed (Chang et al., 2000).

Evolutionary Algorithms (EAs) suit this kind of problem. They evolve a population of candidate solutions through selection, crossover, and mutation, with no gradient information required (Eiben and Smith, 2015), so the mixed combinatorial-continuous structure of cardinality-constrained optimization maps naturally onto real-valued representations. Prior work shows that genetic algorithms find competitive portfolios on standard benchmark instances (Chang et al., 2000; Streichert et al., 2003; Lwin et al., 2014), but these evaluations are in-sample only, use datasets of at most 225 stocks, and ignore transaction costs.

This thesis applies a GA with a cardinality constraint ($K \in [10, 30]$) and weight bounds $[0.02, 0.15]$ to a large- and mid-cap US equity universe of roughly 867 stocks per month, evaluated Out-of-Sample (OOS) over 252 monthly periods (2005–2025) with transaction costs deducted from all reported returns. It is compared against unconstrained MVO, constrained MVO, and the $1/N$ benchmark under a rolling framework (DeMiguel et al., 2009). GA hyperparameters are tuned with Optuna (Akiba et al., 2019) on an initial 2005–2012 window, an overlap discussed in Section 5.6. The work makes two contributions. First, it tests a cardinality-constrained GA out-of-sample at this scale and with transaction costs, a setting prior GA studies have not examined. Second, it shows that an automatically tuned turnover penalty can dominate the fitness function and degrade performance: the tuned $\lambda = 1.8437$ suppresses cardinality toward $\bar{K} = 16.3$ and costs more in foregone return than it saves, as the ablation and fixed-K sweep jointly confirm. The broader point is that in-sample Sharpe on a high-dimensional, noisy covariance estimate is a poor fitness proxy for a sparse out-of-sample portfolio, and penalty tuning can amplify this disconnect rather than correct it.

Research Questions.

RQ1 What is the impact of a 0.15 maximum weight bound on out-of-sample risk-adjusted performance relative to unconstrained MVO?

RQ2 Under realistic portfolio constraints (cardinality $K \in [10, 30]$, weight bounds $[0.02, 0.15]$, and transaction costs $\gamma = 0.3\%$), does the GA achieve out-of-sample Sharpe Ratios (SRs) competitive with or superior to constrained MVO, unconstrained MVO, and the $1/N$ benchmark?

RQ3 How does portfolio cardinality K affect out-of-sample risk-adjusted return, portfolio concentration, and turnover in the GA, and does the adaptive K mechanism achieve performance comparable to fixed-K configurations?

Hypotheses.

H1 Imposing a 0.15 maximum weight bound on MVO improves or preserves out-of-sample Sharpe relative to unconstrained MVO, by reducing sensitivity to estimation error in the covariance matrix (Jagannathan and Ma, 2003).

H2 When subject to realistic portfolio constraints, the GA achieves an out-of-sample SR competitive with or superior to constrained MVO, and superior to unconstrained MVO on downside risk metrics (drawdown, volatility), while remaining competitive with the $1/N$ benchmark (DeMiguel et al., 2009).

H3 Higher cardinality configurations, particularly $K \in \{25, 30\}$, produce higher out-of-sample net Sharpe ratios than the adaptive GA and small fixed-K settings, though the relationship between K and Sharpe is not strictly monotonic.

2 Background

Mean-Variance optimization. Markowitz (1952) cast portfolio selection as a quadratic optimization problem. Given N assets with expected return vector μ and covariance matrix Σ, the objective is to find weights w that minimize portfolio variance for a given target return:

$$\min_{w} \; w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu \geq r^*, \quad \sum_i w_i = 1, \quad w_i \geq 0 \tag{1}$$

The set of optimal solutions across all choices of r^* forms the efficient frontier. For Sharpe ratio maximization, the target return is dropped and the tangency portfolio is found directly. Because Equation 1 is a convex quadratic program, standard solvers such as Sequential Least Squares Programming (SLSQP) can find the solution efficiently.

Estimation Error. The practical difficulty with MVO is that μ and Σ are unknown and must be estimated from historical data. Michaud (1989) identifies what he calls the error maximization problem: the optimizer tends to assign large weights to assets with high estimated returns, which are also the assets whose returns are most likely to be overestimated. Small changes in inputs produce large changes in the optimal portfolio, and out-of-sample performance degrades substantially.

In this study the problem is especially acute. With $N \approx 867$ stocks and a 60-month window, the sample covariance matrix has rank at most $T - 1 = 59$, far below its $N \times N$ dimensions, so the optimizer largely fits sample noise and in-sample quality does not carry over out-of-sample.

Cardinality Constraints. Limiting the number of assets held at once addresses part of this problem. Restricting the portfolio to K stocks, a cardinality constraint, reduces the number of covariance terms that must be estimated accurately and prevents extreme concentration driven by noisy return estimates. The constraint takes the form:

$$K_{\min} \leq |\{i : w_i > 0\}| \leq K_{\max} \tag{2}$$

Adding this constraint makes the problem NP-hard. Chang et al. (2000) therefore tackle it with heuristics (genetic algorithms, tabu search, and simulated annealing) rather than exact solvers. The solver must simultaneously decide which K assets to hold and how much weight to assign each. Standard convex solvers handle the weight allocation given a fixed asset set, but not the asset selection itself.

Evolutionary Algorithms. EAs maintain a population of candidate solutions and improve it iteratively through selection, crossover, and mutation (Eiben and Smith, 2015). No gradient information is required, so EAs can handle the mixed combinatorial-continuous structure of cardinality-constrained portfolio optimization directly.

The GA described in Section 4 follows this framework. Each chromosome encodes a candidate portfolio as a real-valued weight vector, with zero entries representing excluded stocks. Crossover and mutation jointly explore which assets to hold and at what weights, and the repair operator restores feasibility after each step.

3 Related Work

3.1 MVO and Constraints

Markowitz (1952) established the mean-variance framework that both MVO benchmarks in this study implement. The resulting portfolios are optimal only if μ and Σ are known, and both must be estimated from historical data.

Michaud (1989) identifies the central failure. The optimizer assigns large weights to assets with high estimated returns, which tend to be the noisiest, concentrated in stocks with the most extreme historical performance. The resulting portfolio fits historical noise, which is the central reason MVO tends to underperform simpler strategies out-of-sample.

DeMiguel et al. (2009) quantify the scale of this failure. Across seven datasets, the naive $1/N$ strategy matches or outperforms 14 optimized portfolio models on out-of-sample Sharpe ratio, including mean-variance, Bayes-Stein, and factor-based variants. They also show that the estimation window required for optimization to consistently beat equal weighting is far longer than any real dataset provides. Their rolling 60-month evaluation protocol is adopted here, and $1/N$ is included as a benchmark for the same reason: it is a simple strategy that any optimization approach should in principle be able to beat.

Jagannathan and Ma (2003) offer a partial fix. Imposing weight upper bounds on MVO reduces out-of-sample risk, and they show this is equivalent to shrinking the sample covariance toward a structured target. The Constrained MVO benchmark uses a 0.15 upper bound, in the spirit of Jagannathan and Ma (2003), and is kept as a separate baseline to isolate the weight-bound effect from the cardinality effect, a distinction the prior literature does not draw.

3.2 Evolutionary Algorithms for Portfolio Optimization

Chang et al. (2000) formally define the cardinality-constrained portfolio optimization problem and demonstrate that heuristics, including genetic algorithms, find good approximate solutions across five equity datasets. This study adopts their cardinality-constrained formulation. The range $K \in [10, 30]$ follows common practice in later work (Streichert et al., 2003; Lwin et al., 2014). Their evaluation is in-sample only: solutions are assessed against the unconstrained efficient frontier on datasets of at most 225 stocks, with no transaction costs.

Streichert et al. (2003) show that operator-level design choices have a substantial effect on EA performance for this problem class. Real-valued encoding outperforms binary alternatives, repair operators are essential for maintaining feasibility, and Lamarckian local search improves convergence. The GA in Section 4 follows all three design principles. Their evaluation uses the same Library benchmark instances as Chang et al. (2000), again without out-of-sample testing or transaction costs.

Lwin et al. (2014) take the approach further with a learning-guided multi-objective EA that extracts patterns from non-dominated solutions and uses them to bias offspring generation. The result converges faster than standard multi-objective EAs on the same benchmark instances. Out-of-sample evaluation and transaction costs are not included.

All three studies are in-sample only, with no transaction costs. The largest dataset contains 225 stocks, well below the $N \approx 867$ eligible stocks used here. This study evaluates the GA over 252 out-of-sample periods on Center for Research in Security Prices (CRSP) data, with a turnover penalty embedded in the fitness function and transaction costs deducted from all reported returns. A single-objective fitness function is used rather than Pareto optimization, primarily because aggregating results across 252 periods and comparing against MVO benchmarks is more straightforward with a scalar metric.

3.3 Out-of-Sample Evaluation Methodology

The rolling OOS evaluation protocol in this study follows DeMiguel et al. (2009), who apply a 60-month rolling window with monthly rebalancing across multiple datasets and report OOS Sharpe ratio as the primary performance metric. Their study does not include cardinality-constrained strategies or transaction costs, and the datasets used contain at most 25 assets.

The present study adopts the same rolling-window protocol. The choice of a rolling over an expanding window is taken up in Section 5.3.

This study applies the same protocol with two extensions. Transaction costs are deducted at $\gamma = 0.003$ per unit of portfolio turnover, and the GA fitness function includes a turnover penalty during optimization, neither of which appears in DeMiguel et al. (2009). The eligible universe is also considerably larger: $N \approx 867$ stocks versus the 25-asset datasets in their study, which makes the covariance estimation substantially harder and motivated the Ledoit-Wolf shrinkage approach described in Section 5.6.

3.4 EA Parameter Control and Tuning

Eiben and Smith (2015) distinguish between parameter setting, fixed before the run, and parameter control, changed during it. Control strategies range from deterministic schedules to self-adaptive schemes where parameters are encoded in the chromosome and co-evolved with the solution. This study uses externally-tuned static parameters: p_c (crossover probability), p_m (mutation probability), σ_m (Gaussian mutation step size), and λ (turnover penalty weight in the fitness function) are set by Optuna before evaluation begins and do not change during evolution.

Static parameter setting (tuning) was chosen over adaptive control schemes to keep the algorithm's behaviour interpretable and reproducible. Since this study is primarily about OOS portfolio evaluation rather than EA design, the added complexity of self-adaptation is harder to justify. Optuna (Akiba et al., 2019) provides a more systematic approach to parameter selection than manual tuning. Its Tree-structured Parzen Estimator (TPE) sampler concentrates subsequent trials near configurations that performed well in earlier ones, which suited the limited trial budget better than grid or random search. That mattered here: each trial runs the GA over all 96 rebalancing periods in the tuning window, and only 15 trials were run.

4 Algorithm Description

4.1 Representation

Each candidate portfolio is a real-valued vector $w \in \mathbb{R}^N$, where $N \approx 867$ is the number of eligible stocks at rebalancing date t. Exactly K entries are non-zero, each bounded to $[0.02,\ 0.15]$, and the full vector sums to one:

$$\sum_{i=1}^{N} w_i = 1, \qquad w_i \in \{0\} \cup [0.02,\ 0.15], \qquad K_{\min} \le |\{i : w_i > 0\}| \le K_{\max} \tag{3}$$

where $K_{\min} = 10$ and $K_{\max} = 30$ (Chang et al., 2000), and $w_i = 0$ indicates the stock is not held. A non-zero entry encodes both the selection decision and the weight in a single value, with no separate decoding step required.

A binary selection vector paired with a separate weight vector would have to stay consistent through crossover and repair, and Gaussian mutation and weight blending do not map cleanly onto a binary flag. Streichert et al. (2003) find real-valued encoding outperforms binary alternatives on this problem class, and Eiben and Smith (2015) recommend it for continuous weight optimization generally.

K is drawn from $\mathcal{U}\{K_{\min}, \ldots, K_{\max}\}$ at initialisation, so the population covers portfolios of varying size from the start. The repair operator in Section 4.7 keeps K within bounds throughout evolution.

4.2 Fitness Function

At each rebalancing date, the GA maximises:

$$f(w) = \mathrm{SR}(w) - \lambda \cdot \tau(w,\, w_{\text{prev}}) \tag{4}$$

where $\mathrm{SR}(w)$ is the monthly Sharpe ratio computed from the 60-month estimation window:

$$\mathrm{SR}(w) = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \tag{5}$$

with μ the sample mean return vector and Σ the regularised sample covariance matrix. Monthly rather than annualised Sharpe is used because the $\sqrt{12}$ factor cancels in the ratio and does not affect how solutions are ranked. $\tau(w, w_{\text{prev}})$ is the portfolio turnover defined in Equation 13, and $\lambda = 1.8437$ is the Optuna-tuned penalty weight (Akiba et al., 2019).

The turnover penalty discourages high-Sharpe portfolios that require large trades from the previous holdings. Without it, the algorithm ignores trading costs during optimization. At the first rebalancing date no previous portfolio exists, so the penalty is set to zero. This differs from the evaluation convention, where first-period turnover is set to 1.0 to reflect the full cost of constructing the portfolio from scratch.

The fitness is an in-sample quantity: μ and Σ come from the estimation window, not from realised returns. The evaluation metric, net Sharpe reported in Section 6, uses realised out-of-sample returns with $\gamma \cdot \tau_t$ deducted each month.

4.3 Initialisation

Each individual is constructed independently. K is drawn from $\mathcal{U}\{K_{\min}, \ldots, K_{\max}\}$ and K stocks are selected without replacement from the eligible universe. Weights are drawn from the symmetric Dirichlet $\mathrm{Dir}(\mathbf{1}_K)$, which samples the K-asset simplex uniformly and sums to one by construction, so every initial portfolio satisfies the budget constraint without rescaling. The repair operator (Section 4.7) then projects the weights onto $[W_{\min}, W_{\max}]$ while preserving the budget constraint.

Each of the 8 parallel GA runs receives an independent random seed, so initial populations are uncorrelated across runs. Random initialisation follows standard practice (Eiben and Smith, 2015).

4.4 Selection

Tournament selection is used to choose parents for crossover. At each selection event, three individuals are drawn uniformly at random from the population without replacement, and the one with the highest fitness is returned as a parent (Eiben and Smith, 2015). Each crossover uses two parents, so two independent tournaments are run per crossover.

Fitness-proportionate (roulette-wheel) selection is undefined here because Equation 4 can go negative when the penalty exceeds the Sharpe ratio. Tournament selection compares candidates ordinally, so the scale does not matter. With $k = 3$, selection pressure is moderate, enough to drive improvement without collapsing diversity early (Eiben and Smith, 2015).

4.5 Crossover

Crossover fires with probability $p_c = 0.6054$ (Optuna-tuned). Otherwise, both parents are returned unchanged. When it fires, two children are produced by swapping parent roles, extracting two candidate solutions from each tournament pair.

Each child is built from the union of assets held by either parent. K stocks are sampled from this union proportional to the average parental weight:

$$p_i = \frac{w_{1,i} + w_{2,i}}{2}, \quad i \in \text{union} \tag{6}$$

Stocks held by both parents are more likely to appear in the child. Drawing cardinality independently from $\mathcal{U}\{K_{\min}, \ldots, K_{\max}\}$ also means portfolio size varies between children, not just composition.

Weights are blended with a single shared $\alpha \sim \mathcal{U}(0, 1)$:

$$w_{\text{child},i} = \alpha \cdot w_{1,i} + (1 - \alpha) \cdot w_{2,i} \tag{7}$$

for each selected stock i. A single α applies to all selected stocks, so the child's weights shift uniformly toward one parent rather than mixing each stock independently. The repair operator (Section 4.7) restores feasibility after blending.

4.6 Mutation

Two mutation operators are applied via independent Bernoulli draws, each with probability $p_m = 0.1370$ (Optuna-tuned). Both can fire in the same call, only one can fire, or neither. The operator is skipped entirely if both draws fail.

Gaussian perturbation adds $\mathcal{N}(0, \sigma_m)$ noise with $\sigma_m = 0.1469$ to the weights of all currently held stocks. Weights that go negative after perturbation are set to zero, dropping that stock from the portfolio. This can incidentally reduce cardinality below $K_{\min}$, which the repair operator then corrects.

Asset swap removes one held stock by setting its weight to zero and introduces one currently unheld stock at $W_{\min} = 0.02$, keeping the initial change small and leaving the rest to repair. Cardinality is unchanged at this point.

Repair (Section 4.7) is applied once at the end, after both operators have fired, so both perturbations accumulate before constraints are enforced (Eiben and Smith, 2015).

4.7 Repair Operator

Repair is called after crossover, after mutation, and during initialisation. In the mutation operator it fires once at the end, after both sub-operators have had a chance to act (Section 4.6).

The procedure has two stages. First, cardinality is enforced on the raw weights. If $|\{i : w_i > 0\}| < K_{\min}$, zero-weight stocks chosen uniformly at random from the currently unheld set are activated at $W_{\min} = 0.02$ until the minimum is reached.

Second, the non-zero weights are projected onto the bounded simplex:

$$\mathcal{S} = \left\{ w \in \mathbb{R}^K : \sum_{i=1}^{K} w_i = 1, \quad W_{\min} \le w_i \le W_{\max} \right\} \tag{8}$$

The projection finds the smallest uniform shift λ^* that pulls all weights into $[W_{\min}, W_{\max}]$ while keeping their sum at one. Formally, λ^* is found via bisection such that $\sum_i \text{clip}(v_i - \lambda^*, W_{\min}, W_{\max}) = 1$, then $w_i = \text{clip}(v_i - \lambda^*, W_{\min}, W_{\max})$. This enforces the weight bounds and budget constraint in a single pass.

Simply clipping weights to $[W_{\min}, W_{\max}]$ and renormalising is unreliable: it can push other weights back out of bounds, needing several iterations. The bisection projection is feasible in one pass.

After projection, cardinality is checked again. In rare cases the projection sends a weight near $W_{\min}$ to zero, reducing cardinality below $K_{\min}$. If this happens, the function activates a random zero-weight stock at $W_{\min}$ and calls itself once recursively. Recursion depth is capped at one call.

4.8 Local Refinement and Elitism

Elitism. At the end of each generation, the top 5% of individuals by fitness, 5 out of 100, are copied into the next population unchanged (Eiben and Smith, 2015). Their fitness values are carried over rather than recomputed, saving 5 evaluations per generation. The remaining 95 slots are filled with offspring from selection, crossover, and mutation.

Local refinement. Before the best elite is passed to the next generation, it undergoes a short greedy search (Streichert et al., 2003). The procedure runs for up to $L = 5$ iterations. At each step, two stocks are drawn at random from the currently held portfolio, and a fixed weight δ is transferred from the first to the second:

$$w_i \leftarrow w_i - \delta, \qquad w_j \leftarrow w_j + \delta \tag{9}$$

The transfer amount is bounded by:

$$\delta = \min(\delta_0,\ w_i - W_{\min},\ W_{\max} - w_j) \tag{10}$$

where $\delta_0 = 0.01$, so both weights stay within $[W_{\min}, W_{\max}]$ after the shift. The step is kept small relative to the $[0.02, 0.15]$ weight range so each greedy move makes a local adjustment. A larger step would tend to overshoot good allocations and be rejected more often. The candidate is accepted only if fitness improves. Otherwise, the original weights are kept. Since the shift only moves weight between two held stocks, the budget constraint and cardinality remain satisfied. No repair is needed.

Local refinement is Lamarckian: the refined weights replace the original chromosome, so improvements carry forward into the next generation (Streichert et al., 2003). It is applied to the best elite only, not to the full set of elites, to limit the additional computational cost per generation.

4.9 Hyperparameter Tuning with Optuna

The four parameters most affecting GA behaviour, p_c, p_m, σ_m, and λ, are tuned using Optuna (Akiba et al., 2019) before the main evaluation begins. These set the search's exploration–exploitation balance (p_c, p_m, σ_m) and the shape of the fitness function (λ). Population size and generation count are left fixed instead: their effect on in-sample fitness is largely monotonic, so tuning them would mostly push toward the largest setting the compute budget allows rather than an informative optimum, and early stopping already lets each run use only as many generations as it needs. All other parameters are fixed in advance (Table 1).

Each trial runs the GA across all 96 rebalancing periods in the tuning window with reduced computational settings (3 runs per period, 30 generations per run, versus 8/200 in the main evaluation), which keeps each trial well below the cost of a full 252-period evaluation.

The search ranges and selected values are listed in Table 2. At $n_{\text{trials}} = 15$, the budget is small for a four-dimensional search space, and better configurations may exist.

The tuned parameters are fixed for the full 2005–2025 evaluation and do not change between rebalancing periods. Because the tuning window (2005–2012) overlaps the evaluation period, the parameters are not strictly out-of-sample for the first 96 periods. Implications are discussed in Sections 5.6 and 8.3.

5 Experimental Setup

5.1 Data

Stock return data come from the CRSP monthly stock file (`msf_v2`), accessed via Wharton Research Data Services (WRDS), covering January 2000 to December 2025. The data are in CIZ format, the current CRSP standard for US equities. The return variable used throughout is `MthRet`, which already incorporates delisting returns, so no separate adjustment is required. In the rare case of duplicate observations for the same security and month, caused by multiple distribution events, only the first record is retained.

The variables extracted are: the permanent security identifier (`permno`, stable across ticker changes), primary exchange (`primaryexch`), five share-class indicators (`sharetype`, `securitytype`, `securitysubtype`, `usincflg`, `issuertype`) that together flag ordinary common stock, the month-end price (`mthprc`) and shares outstanding (`shrout`), whose product is market value, and the total monthly return (`mthret`, including reinvested dividends). Section 5.2 describes how they filter the eligible universe.

The monthly risk-free rate is taken from FRED, using the 3-Month Treasury Bill secondary market rate (`DTB3`), reported as an annualised percentage. It is converted to a monthly decimal as:

$$r_{f,t} = \frac{\mathtt{DTB3}_t/100}{12} \tag{11}$$

The monthly excess returns are then $r_{e,i,t} = r_{i,t} - r_{f,t}$, where $r_{i,t}$ is `MthRet` for stock i in month t. Excess returns are used throughout rather than raw returns because they remove the time-varying risk-free rate component, making performance comparisons across different interest rate environments more interpretable. Since CRSP dates are end-of-month (e.g., 31 January 2005) while FRED dates are start-of-month (1 January 2005), the two series are matched on the year-month period. The full codebase is publicly available at https://github.com/georgeded/ga-portfolio-optimization.

5.2 Universe Construction

At each monthly rebalancing date t, the eligible investment universe is constructed by applying three sequential filters to the full CRSP dataset described in Section 5.1.

Exchange and share class. Only securities listed on the New York Stock Exchange (NYSE) or National Association of Securities Dealers Automated Quotations (NASDAQ) are retained ($\texttt{primaryexch} \in \{\mathrm{N}, \mathrm{Q}\}$). Common stocks are identified using the CIZ format criteria: `ShareType=NS`, `SecurityType=EQTY`, `SecuritySubType=COM`, `usincflg=Y`, and $\texttt{issuertype} \in \{\mathrm{CORP}, \mathrm{ACOR}\}$. These are the CIZ equivalents of the legacy $\texttt{shrcd} \in \{10, 11\}$ screen used in the older SIZ format.

Market capitalisation. Stocks with a market cap below \$2 billion are excluded to avoid illiquid micro-cap securities. Market cap is computed as:

$$\mathrm{MktCap}_{i,t} = |P_{i,t}| \times \texttt{shrout}_{i,t} \times 1{,}000 \tag{12}$$

where $P_{i,t}$ is the end-of-month price and `shrout` is shares outstanding in thousands of shares, so the product is in dollars. To avoid look-ahead bias, the filter uses each stock's market cap from the month immediately preceding t (the last month-end observation before the rebalancing date), so only information available at t enters the screen.

Return history. Each stock must have exactly 60 non-missing monthly returns in the estimation window $[t-60,\, t-1]$. This ensures the sample covariance matrix can be fully estimated without gaps, and is consistent with the 60-month window used in all optimization models.

Together, these filters yield an average eligible universe of approximately 867 stocks per month, ranging from 487 to 1,105 across the 252 rebalancing periods. The first eligible rebalancing date is January 2005, as the 60-month history requirement consumes the initial data from January 2000. The shared estimation function used by all models enforces the same completeness criterion internally, dropping any stock with a missing return within the 60-month window before computing μ and Σ. Because both filters require exactly 60 complete monthly returns, the optimization universe closely matches the eligible universe at each period.

5.3 Evaluation Protocol

The evaluation follows DeMiguel et al. (2009): a rolling OOS setup with a fixed 60-month estimation window and monthly rebalancing over 252 periods (January 2005–December 2025).

This study uses a rolling rather than expanding window because the evaluation period spans several structural breaks, the dot-com recovery, the 2008 financial crisis, and the COVID-19 shock. Mixing data from these very different regimes in one estimation window can distort the covariance matrix (López de Prado, 2018).

At each rebalancing date t, each model computes a target weight vector w_t. Between rebalancing dates, weights drift as stock prices move, so the pre-rebalance weight vector $\tilde{w}_t$

differs from the previous period's target. Monthly portfolio turnover is computed using these drifted weights:

$$\tau_t = \frac{1}{2}\sum_{i=1}^{N} |w_{i,t} - \tilde{w}_{i,t}| \tag{13}$$

Transaction costs are then deducted per unit of portfolio turnover following DeMiguel et al. (2009). The baseline rate $\gamma = 0.003$ is an assumption of this study, varied in Appendix C, giving a net monthly excess return of $r_{e,t}^{\text{net}} = r_{e,t} - \gamma \cdot \tau_t$. All performance metrics reported in Section 6 are based on net returns unless stated otherwise.

Because the GA is stochastic, multiple independent runs are executed at each rebalancing date. The run whose in-sample fitness is closest to the median fitness across all runs is selected as the canonical run (ties broken by index order). Its portfolio weights are used for evaluation. The number of parallel runners and the rationale for median selection are detailed in Section 5.6.

The primary metric is the net annualised SR (Sharpe, 1994). Secondary metrics reported are: annualised return, annualised volatility, Sortino ratio (Sortino and Price, 1994), maximum drawdown, average portfolio turnover, and Herfindahl-Hirschman Index (HHI) concentration.

Pairwise Sharpe ratio differences are tested using the Jobson-Korkie test (Jobson and Korkie, 1981), with the correction from Memmel (2003). As a robustness check, a paired two-sided t-test on monthly net excess returns is also reported.

5.4 Benchmarks

Three benchmarks are evaluated alongside the GA. These cover three quite different approaches: full optimization, optimization with a weight cap, and no optimization at all.

Unconstrained MVO maximises the Sharpe ratio subject only to long-only constraints (weight bounds $[0, 1]$), solved via SLSQP (DeMiguel et al., 2009). This is the unconstrained mean-variance optimum given the sample estimates, and is the classical optimization baseline. If imposing constraints improves out-of-sample performance, unconstrained MVO may underperform relative to more restricted strategies. That comparison is what **RQ1** (Section 1) addresses.

Constrained MVO applies the same Sharpe maximisation but limits individual weights to $[0, 0.15]$, following Jagannathan and Ma (2003), who show that weight upper bounds reduce out-of-sample risk by acting as implicit covariance shrinkage. This benchmark shares the GA's upper weight bound of 0.15 but invests in all eligible stocks rather than a cardinality-limited subset. Since both share the 0.15 upper weight bound, differences between this benchmark and the GA isolate the effect of the cardinality constraint rather than the weight cap, which is the comparison **RQ2** addresses.

$1/N$ assigns equal weight $1/N$ to all N eligible stocks at each rebalancing date and requires no parameter estimation (DeMiguel et al., 2009). DeMiguel et al. (2009) show that $1/N$ is surprisingly hard to beat out-of-sample, making it a natural reference point for any optimized strategy. Whether the GA can consistently outperform $1/N$ is a central part of **RQ2** (Section 1).

Why these benchmarks. Together they separate the weight-bound effect (**RQ1**) from the cardinality effect (**RQ2**). Both MVO variants are continuous long-only Sharpe-maximisation problems solved with SLSQP. Adding cardinality makes the problem combinatorially hard (Section 2), which is why a convex solver cannot address the full problem.

Alternatives considered. The Bayes-Stein shrinkage estimator (DeMiguel et al., 2009) addresses estimation error through return shrinkage rather than constraints, and including it would conflate the constraint effect with better return estimation. The minimum variance portfolio was excluded because it ignores expected returns entirely and does not share the GA's Sharpe-maximisation objective. Factor-model-based strategies were excluded as they introduce a separate return-modelling step that is outside the scope of this study.

All three benchmarks share the same eligible universe, 60-month estimation window, and transaction cost model ($\gamma = 0.003$) as the GA, so differences in performance should reflect portfolio construction choices rather than differences in data or cost assumptions. The SLSQP solver uses three random restarts per period. The best result across restarts is returned. The near-identical results across both MVO variants suggest solver convergence is not a material concern at this universe size.

5.5 Parameter Settings

The GA parameters fall into two groups. The first set is fixed in advance, based on standard GA practice and prior literature, as listed in Table 1. The second set, shown in Table 2, is tuned with Optuna on the 2005–2012 period. The procedure is described in Section 4.9.

Table 1: Fixed GA parameters used in all experiments.

Parameter	Value	Source
Population size (μ)	100	Standard GA practice**
Max generations	200	Standard GA practice**
Elite fraction	5%	Standard GA practice**
Tournament size	3	Standard GA practice**
Early stopping	20 stagnant generations	Standard GA practice**
Cardinality K	$[10, 30]$ adaptive	Chang et al. (2000)
Weight bounds	$[0.02, 0.15]$	Upper: Jagannathan and Ma (2003), lower: design choice
Transaction cost γ	0.003	Design choice, varied in App. C
Local refinement steps	5	Streichert et al. (2003)
Parallel runners	8	Hardware constraint*

*See Section 5.6.

**These values follow standard GA practice, Eiben and Smith (2015) covers the operators and their typical ranges but does not prescribe these exact values.

Table 2: Optuna-tuned GA parameters (Akiba et al., 2019). Tuning period: 2005–2012 $n_{\text{trials}} = 15$. The *Default* column lists conventional EA values (Eiben and Smith, 2015) used as the convergence baseline in Appendix A.

Parameter	Search range	Tuned value	Default
p_c (crossover rate)	$[0.60, 0.95]$	0.6054	0.8
p_m (mutation rate)	$[0.01, 0.30]$	0.1370	0.1
σ_m (mutation step)	$[0.01, 0.15]$	0.1469	0.05
λ (turnover penalty)	$[0.0,\ 2.0]$	1.8437	0.0

The cardinality sensitivity analysis (RQ3) evaluates five fixed-K configurations: $K \in \{10, 15, 20, 25, 30\}$. Each run uses the same Optuna-tuned parameters from Table 2, the same 8 parallel runners, and the same 200 generations per period as the main adaptive-K evaluation (Section 5.3). The only difference is that cardinality is fixed at the target value rather than drawn from $\mathcal{U}\{K_{\min}, \ldots, K_{\max}\}$. Both $K_{\min}$ and $K_{\max}$ are set to K so the repair operator enforces the fixed cardinality exactly. All five configurations are evaluated across all 252 out-of-sample periods (January 2005–December 2025).

5.6 Methodological Deviations

Three deviations from the original methodology are documented:

1. **Parallel runners.** The methodology specifies 30 parallel GA runs per rebalancing period (aggregated by median). The cloud VM (Google Cloud `c2-standard-16`, 16 vCPUs over 8 physical cores) supported only 8 runners, one per physical core to avoid hyperthreading contention on the fitness evaluations. Scaling to 30 was not feasible on this hardware. The main GA evaluation (252 periods, 8 runners) ran in about 35 minutes, and the full pipeline (Optuna tuning, MVO, fixed-K, and the λ-ablation) in about 13 hours, roughly 8 of them MVO. The canonical run is the one whose in-sample fitness is closest to the median across the 8 runs. With 8 rather than 30, it carries slightly more stochastic noise. This affects only the GA results, not the benchmark comparisons.

2. **Covariance estimation.** The sample covariance matrix $\Sigma \in \mathbb{R}^{N \times N}$ (with $N \approx 867$) is rank-deficient when estimated from $T = 60$ observations ($T \ll N$, ratio $N/T \approx 14.5$). To address this, the analytical Ledoit-Wolf shrinkage estimator (Ledoit and Wolf, 2004) is applied via `sklearn.covariance.LedoitWolf`, which computes the optimal shrinkage intensity analytically rather than through cross-validation. This is applied consistently across all models (GA, both MVO variants, and fixed-K runs) through the shared `get_estimation_window` function.

3. **Optuna tuning period overlap.** The GA hyperparameters are tuned on 2005–2012 (96 periods), which is contained within the full evaluation period of 2005–2025 (252 periods). The tuned parameters are therefore not strictly out-of-sample for the first 96 evaluation periods. A non-overlapping tuning window would have left too few periods for either tuning or evaluation, so it was accepted as a practical compromise.

The implications are discussed in Section 8.3. The tuned parameter values and trial metadata are archived in `results/optuna/best_params.json` in the public repository.

6 Experimental Results

6.1 Main Performance Comparison

Table 3 reports net-of-cost performance metrics for all four strategies across 252 monthly periods (January 2005–December 2025). Net annualised Sharpe ratio (Sharpe, 1994), computed on monthly net excess returns after deducting $\gamma \cdot \tau_t$ each period, is the primary metric. The Sortino ratio (Sortino and Price, 1994) shares the same numerator but divides by the downside deviation, the root mean square of negative monthly net excess returns, rather than the total volatility. The annualised return multiplies the sample mean of monthly net excess returns by 12. The annualised volatility scales the sample standard deviation by $\sqrt{12}$. Maximum drawdown is the largest peak-to-trough decline in cumulative net portfolio value during the evaluation period. The average monthly turnover and average transaction cost are the time-series means of τ_t and $\gamma \cdot \tau_t$ for all 252 periods.

Table 3: Main performance comparison, net of transaction costs, January 2005–December 2025. All metrics are computed on monthly net excess returns. Transaction cost is $\gamma = 0.3\%$ per unit traded.

Metric	GA	Constrained MVO	Unconstrained MVO	$1/N$
Ann. Return	5.46%	8.28%	8.28%	9.13%
Ann. Vol.	19.92%	14.26%	14.26%	17.20%
Sharpe	0.2741	0.5810	0.5809	0.5307
Sortino	0.2726	0.5958	0.5958	0.4925
Max DD	-55.76%	-43.07%	-43.07%	-51.31%
Turnover	22.19%	17.07%	17.07%	4.50%
Avg Cost	0.0666%	0.0512%	0.0512%	0.0135%

The GA achieves a net Sharpe ratio of 0.2741 and an annualised net return of 5.46%, below all three benchmarks. Its maximum drawdown (-55.76%) is deeper than that of any benchmark. Average monthly turnover (22.19%) exceeds both MVO variants (17.07%). Constrained and unconstrained MVO produce near-identical results across all reported metrics, with net Sharpe ratios of 0.5810 and 0.5809. Section 6.2 reports significance tests.

6.2 Statistical Significance

Table 4 reports pairwise significance tests comparing the GA against each benchmark. The paired t-test assesses whether mean monthly net excess returns differ. The Jobson-Korkie test (Jobson and Korkie, 1981), corrected by Memmel (2003), tests whether Sharpe ratios differ. The null hypothesis for each comparison is $H_0\colon \mathrm{SR}_A = \mathrm{SR}_B$.

Table 4: Statistical significance of GA performance differences against benchmark strategies, January 2005–December 2025 ($T = 252$). The paired t-test evaluates whether mean return differences equal zero. The Jobson–Korkie test with Memmel correction evaluates whether Sharpe ratios differ.

Metric	GA vs $1/N$	GA vs Unconstrained MVO	GA vs Constrained MVO
Mean Diff. (ann.)	-3.67%	-2.82%	-2.82%
t-stat	-1.139	-1.271	-1.271
t p-value	0.2558	0.2051	0.2051
t significant?	no	no	no
Sharpe Diff.	-0.2566	-0.3068	-0.3069
z-stat	-1.508	-2.843	-2.843
JK p-value	0.1316	0.0045	0.0045
JK significant?	no	yes	yes

The GA's net Sharpe ratio of 0.2741 falls below all three benchmarks. Against both MVO variants the Sharpe underperformance is statistically significant under the Jobson-Korkie test ($p < 0.01$). The paired t-test does not reject ($p = 0.205$ vs constrained, $p = 0.205$ vs unconstrained), indicating the mean return shortfall is not significant at conventional levels. Against $1/N$, neither test rejects (t-test $p = 0.256$, Jobson-Korkie $p = 0.132$).

6.3 Portfolio Characteristics

Table 5 reports structural characteristics of each strategy averaged over the 252 evaluation periods. Avg K is the mean number of stocks with non-zero weight per month. For $1/N$ this equals the full eligible universe, averaging 869.6 stocks per period. For both MVO variants, Avg K reflects the number of non-zero positions returned by SLSQP: approximately 62 on average, despite no cardinality constraint being imposed. Avg HHI is the time-series mean of $\sum_i w_i^2$, where values closer to the theoretical minimum $1/K$ indicate more uniform weighting within the held subset.

Table 5: Portfolio characteristics across strategies, January 2005–December 2025. Avg K is the average number of stocks held per period. HHI is the Herfindahl–Hirschman Index. Transaction cost is $\gamma = 0.3\%$ per unit traded.

Metric	GA	Constrained MVO	Unconstrained MVO	$1/N$
Avg K	16.3	62.0	62.3	869.6
Avg HHI	0.086581	0.033922	0.033923	0.001176
Turnover	22.19%	17.07%	17.07%	4.50%
Avg Cost	0.0666%	0.0512%	0.0512%	0.0135%

The GA holds 16.3 stocks per period on average, far more concentrated than the MVO variants' ≈ 62 and $1/N$'s full universe. Its turnover (22.19%) also exceeds both MVO variants (17.07%). The $1/N$ benchmark has the lowest HHI (0.0012), near its theoretical minimum by construction. Cross-strategy HHI comparison is limited because the minimum $1/K$ differs by portfolio size (Section 6.7).

6.4 Cumulative Returns

Figure 1 plots the growth of \$1 invested in each strategy at the start of January 2005. All four strategies show visible drawdowns during the 2008–09 financial crisis and the COVID-19 shock in early 2020. The cumulative series diverge over the evaluation period. By December 2025, both MVO variants and $1/N$ reach higher terminal values than the GA.

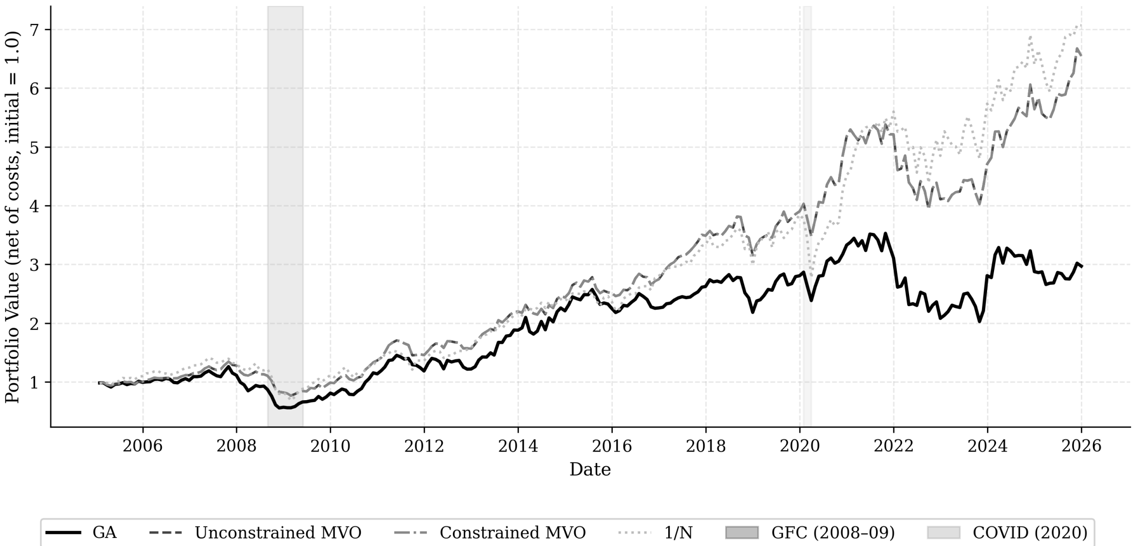

Figure 1: Cumulative net returns of all four strategies (2005–2025). Returns are net of transaction costs ($\gamma = 0.3\%$ per unit traded). Starting value normalised to 1.0 at January 2005.

6.5 Rolling Sharpe Ratio

Figure 2 shows the **12-month** rolling net Sharpe ratio for each strategy from January 2008 to December 2025. All strategies record negative rolling Sharpe ratios during the 2008–09 crisis period. In most sub-periods the GA's rolling Sharpe ratio falls below the MVO variants, with the exception of the 2010–2019 recovery period. All strategies show increased variability around the COVID-19 shock in early 2020.

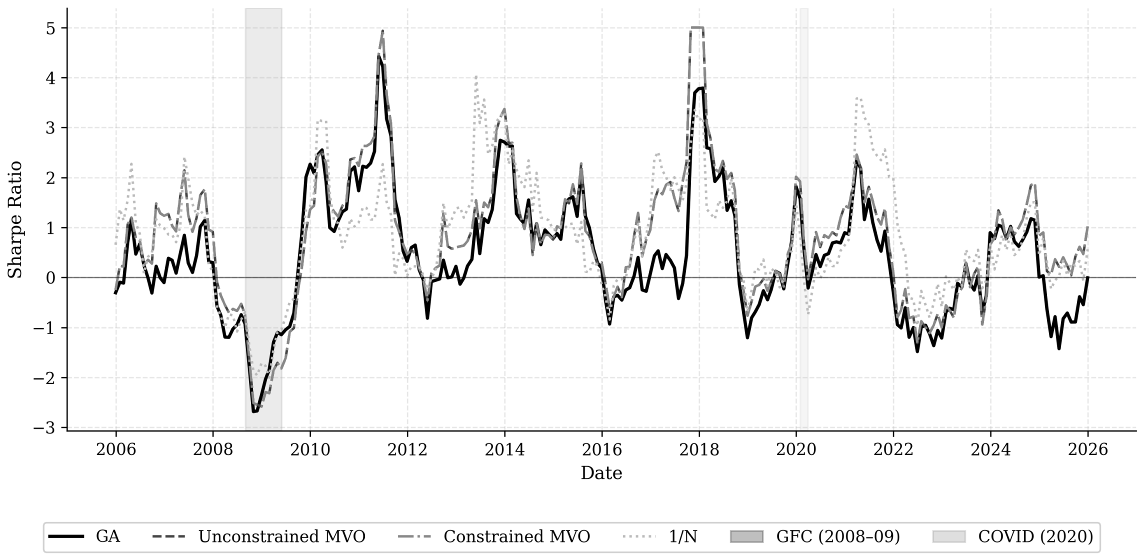

Figure 2: Rolling **12-month** net SR (Sharpe, 1994) for all strategies (2008–2025).

6.6 Turnover and Transaction Costs

Figure 3 plots monthly portfolio turnover for each strategy over the full evaluation period. Shaded bands span ±1 standard deviation. The GA runs at higher turnover than the MVO variants throughout the period. The $1/N$ strategy has the lowest and most stable turnover of the four strategies.

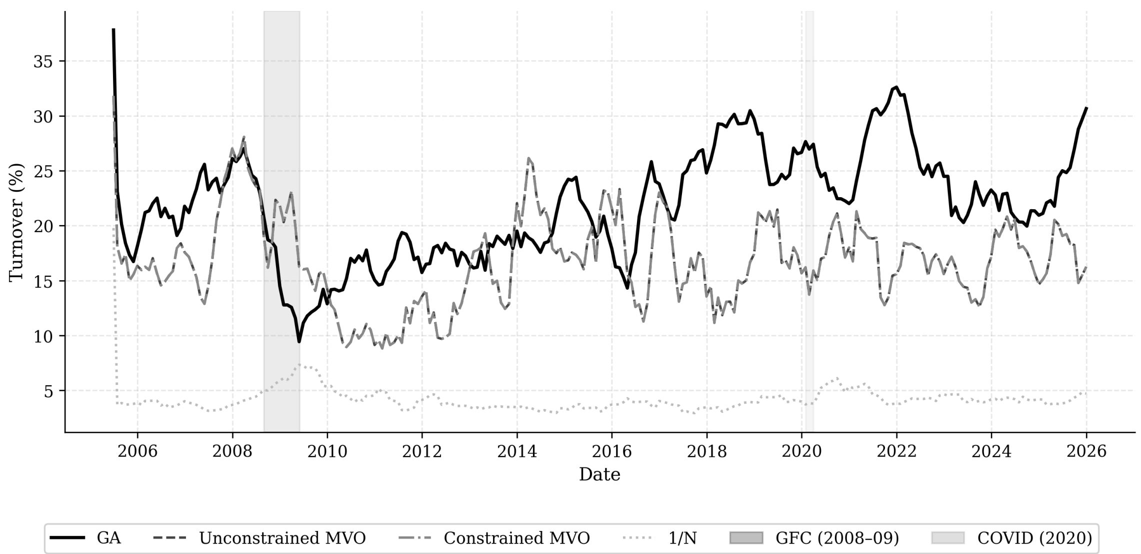

Figure 3: Monthly portfolio turnover for all strategies (2005–2025). Shaded bands indicate ±1 standard deviation.

6.7 Portfolio Concentration

Figure 4 shows portfolio concentration and the GA's adaptive cardinality over time. The GA's HHI sits well above the benchmarks, reflecting its $\approx$ 16-stock portfolio. Because the theoretical minimum $1/K$ differs across strategies ($1/16 \approx 0.063$ for the GA, $1/62 \approx 0.016$ for MVO, $1/867 \approx 0.001$ for $1/N$), the levels are not directly comparable. The adaptive cardinality stays within $[10, 30]$, mean 16.3.

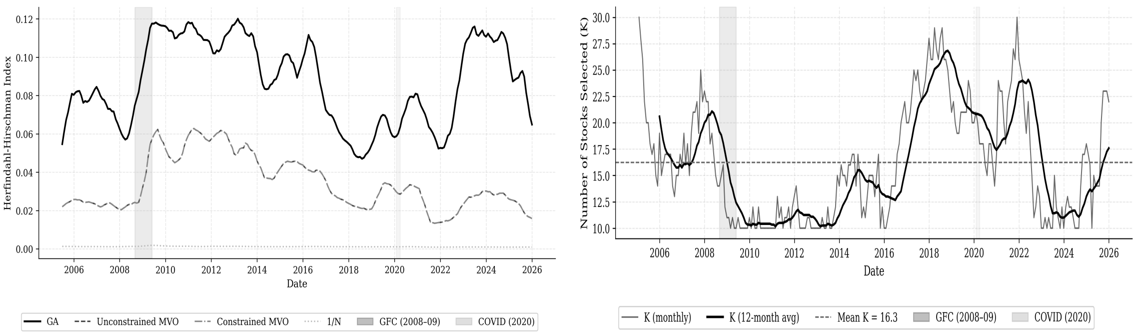

Figure 4: Portfolio concentration and adaptive cardinality over the 2005–2025 evaluation period. Left: HHI concentration. Right: adaptive cardinality K.

6.8 Cardinality Sensitivity

Table 6 reports performance for the five fixed-K configurations across all 252 periods, and Figure 5 plots net Sharpe, turnover, and concentration against K.

Table 6: K-sensitivity analysis: GA performance and portfolio characteristics with fixed cardinality $K \in \{10, 15, 20, 25, 30\}$, January 2005–December 2025. Fixed hyperparameters are used throughout to isolate the effect of K.

Metric	$K = 10$	$K = 15$	$K = 20$	$K = 25$	$K = 30$
Sharpe	0.2632	0.3254	0.3051	0.4400	0.4363
Ann. Return	5.52%	6.28%	5.70%	7.99%	7.35%
Ann. Vol.	20.99%	19.28%	18.68%	18.15%	16.84%
Max DD	-45.74%	-54.64%	-57.93%	-54.58%	-50.19%
Turnover	23.24%	20.21%	19.01%	18.69%	19.59%
Avg Cost	0.0697%	0.0606%	0.0570%	0.0561%	0.0588%
HHI	0.115482	0.088479	0.071612	0.056699	0.043961
Avg K actual	10.0	15.0	20.0	25.0	30.0

7 Analysis and Discussion

7.1 RQ1: Impact of Constraints on Out-of-Sample Performance

Constrained and unconstrained MVO achieve net Sharpe ratios of 0.5810 and 0.5809. H1 is weakly supported: the constrained variant does not underperform, and the difference

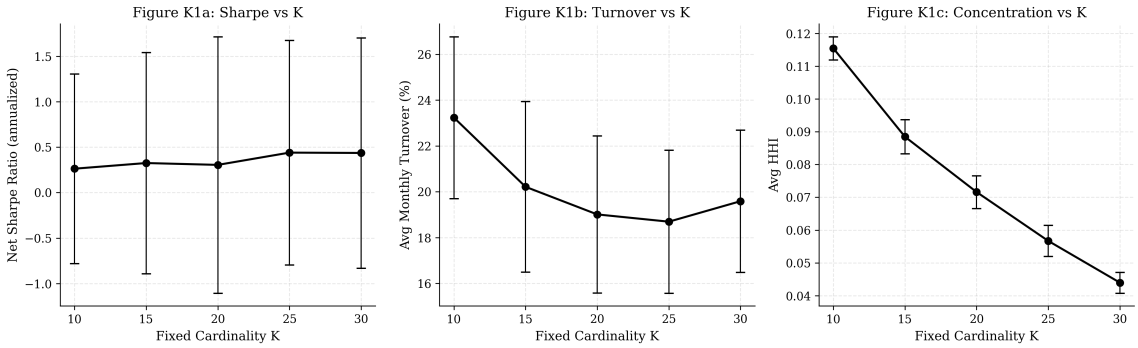

Figure 5: Effect of fixed cardinality K on net Sharpe ratio, average monthly turnover, and HHI concentration (January 2005–December 2025), using the tuned hyperparameters from Table 2. Error bars show one standard deviation of the rolling 12-month estimates.

of 0.0001 is negligible in practice. The likely reason is that the 0.15 upper weight cap rarely binds in a universe of 867 stocks. Jagannathan and Ma (2003) show that weight upper bounds act as implicit covariance shrinkage, but this only has a real effect when the unconstrained optimizer would assign large weights to a small number of stocks. With weight distributed across hundreds of assets, most individual positions are already well below 0.15, so the constraint changes little.

With $N/T \approx 14.5$ (867 stocks, 60-month window), estimation error dominates any benefit from the weight cap. Michaud (1989) describes how MVO amplifies noise in estimated returns rather than genuine signal, which is the main reason the approach tends to underperform simpler strategies out-of-sample. The sample covariance matrix is severely rank-deficient at this ratio, and a simple weight cap is not enough to compensate for the noise it introduces. Both MVO variants end up around 0.581, only marginally above $1/N$ at 0.531, consistent with DeMiguel et al. (2009). Even without a cardinality constraint, SLSQP self-concentrates to ≈ 62 non-zero positions on the near-singular covariance, sparser than $1/N$ but well short of the GA's ≈ 16.

7.2 RQ2: GA vs Benchmarks

The GA falls below all three benchmarks (Table 3). H2 is not confirmed. The Sharpe underperformance relative to MVO is statistically significant under the Jobson-Korkie test ($p < 0.01$), but not under the paired t-test. Neither test rejects against $1/N$ (Table 4).

The primary explanation is an over-aggressive turnover penalty. A full ablation across all 252 evaluation periods with $\lambda = 0$ versus the tuned $\lambda = 1.8437$ shows that removing the penalty raises the GA's net Sharpe from 0.2741 to 0.4993 (Appendix D). The penalty reduces average monthly turnover from 57.1% to 22.2% and average cardinality from $\bar{K} = 25.1$ to $\bar{K} = 16.3$, but the cost in foregone return outweighs the transaction cost saving. $\lambda = 1.8437$ was tuned on 2005–2012 to maximise aggregate net Sharpe over 96 periods. Applied across the full 2005–2025 window, it proves too conservative. During tuning, a high λ visibly reduces trading costs while the out-of-sample return cost of lower cardinality is invisible to the tuner, so Optuna overfit the tuning window's cost structure rather than its return opportunity. The fixed-K analysis in Section 7.3 reaches the same conclusion independently. The adaptive mechanism gravitates toward $K \approx 16$ because the penalty makes smaller portfolios cheaper to maintain. Configurations at $K \in \{25, 30\}$ substantially outperform it under the same tuned parameters.

Removing the penalty does not close the gap completely. At $\lambda = 0$ the GA reaches 0.4993, still short of the 0.581 that both MVO variants achieve. The penalty accounts for most of the shortfall against MVO, about 0.225 of the 0.307 Sharpe-point difference, but roughly 0.08 remains. The fixed-K sweep points to the same conclusion: even the best configuration tops out at 0.44, roughly 0.14 below MVO.

What is left is mostly noise in the in-sample fitness. The covariance is estimated from 60 monthly observations across roughly 867 stocks, so $N/T \approx 14.5$ and the matrix is badly rank-deficient. An in-sample Sharpe built on that estimate says little about how a 10- to 30-stock portfolio will perform out-of-sample. Running the search as 8 independent populations adds further variation, which the deterministic SLSQP solver does not face. The one period where the GA closes most of the gap is the 2010–2019 recovery, when it reaches 0.79. With returns that high and sustained, almost any reasonable portfolio does well and the penalty matters less.

7.3 RQ3: Effect of Cardinality on GA Performance

Table 6 and Figure 5 show how net Sharpe ratio, turnover, and concentration vary across $K \in \{10, 15, 20, 25, 30\}$. H3 is largely confirmed: larger fixed-K configurations outperform the adaptive GA, and the highest Sharpe ratios are reached at $K = 25$ and $K = 30$.

The adaptive GA gravitates toward a mean cardinality of 16.3 stocks per period (Table 5), well below the midpoint of the $[10, 30]$ range. This is a direct consequence of the turnover penalty. Holding fewer stocks makes it easier to keep portfolio changes small between periods, since each rebalancing only needs to touch a small subset of assets. The fitness function therefore creates implicit pressure toward lower K, even when higher cardinality would improve the gross Sharpe ratio.

The fixed-K results make this cost visible. Sharpe ratio rises from $K = 10$ to $K = 25$–30, then levels off or dips slightly, consistent with the non-monotonic relationship stated in H3. Turnover also increases with K, since larger portfolios require more trades to maintain weight bounds after each period's price moves. The concentration metric (HHI) falls monotonically as K increases, as expected.

The gap between the adaptive GA and the best fixed-K configurations suggests the turnover penalty is suppressing cardinality in a way that is costly in risk-adjusted terms. The adaptive mechanism optimizes the wrong objective: it penalises turnover so heavily that the algorithm sacrifices diversification to avoid rebalancing costs. Whether a less aggressive penalty would allow higher cardinality to emerge while still controlling costs remains to be tested. Replacing the weight-change penalty with one based on the number of positions held is one candidate approach, taken up in Section 9.2.

The Optuna-tuned parameters used throughout this analysis ($p_c = 0.6054$, $p_m = 0.1370$, $\sigma_m = 0.1469$, $\lambda = 1.8437$) were fixed identically across all fixed-K runs, isolating cardinality as the only varying factor. Convergence behaviour under tuned versus default parameters is shown separately in Appendix A as a supplementary finding.

8 Threats to Validity

8.1 Internal Validity

The Ledoit-Wolf shrinkage estimator (Ledoit and Wolf, 2004) is applied to address the rank-deficiency of the sample covariance matrix ($N/T \approx 14.5$). The shrinkage intensity is determined analytically from the same 60-month estimation window used to compute μ, so it is itself a sample estimate and carries its own uncertainty. In unusual market regimes, such as the 2008 crisis or the COVID-19 shock, the sample may not be representative enough for the shrinkage intensity to be well-calibrated. Since the same estimator is applied consistently to all models (GA, both MVO variants, and fixed-K runs), relative rankings between strategies are unlikely to change under alternative shrinkage choices, though absolute performance levels may differ.

The MVO solver tolerances were relaxed from the defaults (`ftol`$=10^{-9}$, `maxiter`$=1000$) to `ftol`$=10^{-6}$ and `maxiter`$=200$ to make the optimization tractable at $N \approx 867$. At the original settings, the solver occasionally failed to converge within a reasonable number of iterations. The relaxed tolerances may mean some MVO solutions fall slightly below what tighter settings would find.

The GA ran 8 parallel runners rather than the 30 specified (Section 5.6), so the reported performance carries slightly more stochastic noise than intended. The per-period spread of gross returns (`gross_ret_std`, `gross_ret_iqr` in the output file) shows non-zero dispersion across most periods, consistent with the runs exploring different regions of the search space.

The implementation is validated by 105 unit and integration tests in the repository's `tests/` directory: 38 known-answer tests for the evaluation metrics, 58 for the GA components (repair, simplex projection, initialisation, fitness, selection, crossover, mutation, local refinement, and full runs), and 9 integration tests for drift-weight computation, universe construction, estimation-window correctness, and MVO solver fallback.

8.2 External Validity

The results are based on NYSE and NASDAQ-listed US large- and mid-cap stocks with market capitalisation above $2 billion. How the findings generalise to other settings is unclear. Small-cap stocks tend to be less liquid and have wider bid-ask spreads, so the flat transaction cost model ($\gamma = 0.003$) used here would likely underestimate real costs in that segment, and the cardinality constraint may behave differently when the eligible universe is smaller and noisier. International equities introduce currency risk and different correlation structures that the GA was not designed to account for. In multi-asset settings (bonds, commodities, alternatives), the covariance structure looks quite different, and whether cardinality constraints still help in such settings is unclear. These are limitations on scope, not on the validity of the internal comparisons.

The evaluation period (January 2005–December 2025) spans several high-volatility episodes: the tail end of the dot-com period, the 2008 Global Financial Crisis, and the COVID-19 shock in early 2020. These are not representative of average market conditions. The GA's adaptive cardinality, selecting between 10 and 30 stocks per period, may work particularly well during volatile regimes, when concentrating on a smaller subset of assets could reduce exposure to market-wide noise. In calmer, low-volatility periods the benefit of cardinality constraints over equal weighting or MVO might be smaller. Some of the performance may depend on the unusually volatile period covered here.

8.3 Construct Validity

Net SR (Sharpe, 1994) is used as the primary metric because it is standard in the portfolio evaluation literature and directly comparable with DeMiguel et al. (2009). It assumes returns are approximately normally distributed and treats upside and downside volatility symmetrically, neither of which holds exactly for equity returns, which tend to have fat tails and negative skewness, particularly around crisis periods. The Sortino ratio (Sortino and Price, 1994) is reported alongside it as a robustness check, since it penalises only downside deviation. The GA's Sortino ratio (0.2726) is the lowest of the four strategies, consistent with the Sharpe ranking.

The transaction cost model deducts a flat $\gamma = 0.003$ per unit of portfolio turnover. The proportional-cost form follows DeMiguel et al. (2009), and the rate is this study's assumption. This does not account for market impact, bid-ask spread variation across stocks or time periods, or liquidity differences between positions. In practice, costs depend on trade size and prevailing market conditions. For concentrated positions like the GA's, with individual weights up to 15%, trades could be large enough relative to average daily volume that market impact is non-negligible. Real post-cost performance could therefore be lower than what is reported here, particularly for the GA during periods where the portfolio changes substantially. Sensitivity analysis across five cost levels ($\gamma \in \{0.001, 0.002, 0.003, 0.005, 0.010\}$) confirms the GA's underperformance is robust to the cost assumption. See Appendix C.

Because the tuning window (2005–2012) overlaps the evaluation period, the tuned hyperparameters are not strictly out-of-sample for the first 96 periods, a mild data-snooping risk (López de Prado, 2018). Tuning targets aggregate performance over 96 periods rather than individual dates, which limits the effect. Section 9.1 discusses the walk-forward alternative.

When a stock leaves the eligible universe between rebalancing dates, its position is sold and the capital redeployed. The turnover calculation reindexes the previous (drifted) weights onto the current universe without renormalising, so the weight in exited names reappears as the purchases needed to rebuild the portfolio. The same handling applies to all four strategies. It is not exact: because exited names are dropped rather than carried at their drifted weight, τ_t understates true one-sided turnover by about half the exited weight each period. The effect is small (few stocks exit in any month) and applies equally across strategies, so the relative comparison is unaffected.

8.4 Conclusion Validity

Both the paired t-test and the Jobson-Korkie test (Jobson and Korkie, 1981; Memmel, 2003) are reported: the t-test checks whether mean monthly net returns differ. The Jobson-Korkie test compares Sharpe ratios directly. For the GA against both MVO variants, the Jobson-Korkie test rejects at $\alpha = 0.05$ ($p = 0.0045$ for both), indicating the Sharpe underperformance is statistically distinguishable from zero. The paired t-test does not reject ($p = 0.205$ vs constrained, $p = 0.205$ vs unconstrained), indicating the mean return shortfall is not significant at conventional levels. Against $1/N$, neither test rejects (t-test $p = 0.256$, Jobson-Korkie $p = 0.132$). The GA's Sharpe underperformance relative to MVO is statistically significant under the Jobson-Korkie test, but the return shortfall alone does not reach significance. A longer evaluation period would increase power and could sharpen these conclusions.

The Optuna tuning ran only $n_{\text{trials}} = 15$ in a four-dimensional space. The TPE sampler helps focus the budget (Akiba et al., 2019), but 15 trials is too few to rule out better settings, and whether the tuned configuration is genuinely good or merely reasonable cannot be told from these results. A larger budget would reduce this uncertainty but was not feasible (Section 5.6).

9 Conclusions

This thesis evaluated a GA with cardinality and weight constraints against unconstrained MVO, constrained MVO, and $1/N$ over 252 monthly periods (2005–2025) on a mid- and large-cap US equity universe. The GA underperformed all three on net Sharpe ratio. The main cause is an over-aggressive turnover penalty: removing it lifts net Sharpe from 0.2741 to 0.4993. The fixed-K sweep backs this up independently, with $K \in \{25, 30\}$ well above the adaptive mechanism's $\bar{K} = 16.3$. High-dimensional covariance-estimation noise ($N/T \approx 14.5$), which Ledoit-Wolf shrinkage reduces but does not remove, is a secondary drag, since even at $\lambda = 0$ the GA stays below MVO's 0.581. The underlying issue is that in-sample Sharpe on a noisy, high-dimensional covariance estimate is a poor proxy for out-of-sample performance on a sparse portfolio. A penalty tuned on the same window amplifies this disconnect rather than correcting for it.

RQ1. The 0.15 weight cap barely matters: constrained and unconstrained MVO reach 0.5810 and 0.5809, and both beat $1/N$ (0.5307) only marginally (DeMiguel et al., 2009). H1 holds only weakly, since across 867 stocks the cap rarely binds. SLSQP already self-concentrates to about 62 stocks without any cardinality constraint, far short of where the cap would bite.

RQ2. H2 is not confirmed. The GA (0.2741) trails all three benchmarks, significantly so against MVO under the Jobson-Korkie test but not the paired t-test. The shortfall is driven mainly by the penalty, not the search itself: under the same parameters the fixed-K runs reach 0.44, so the limitation lies in the cardinality the penalty drives the GA toward (Section 7.2).

RQ3. Net Sharpe rises with K up to $K \in \{25, 30\}$, and the adaptive GA sits below every fixed-K setting except $K = 10$. H3 is largely confirmed, though the relationship is not strictly monotonic. The adaptive mechanism settles near $\bar{K} = 16.3$ because the penalty makes small portfolios cheaper to hold (Section 7.3).

Overall, the results do not support the idea that pairing cardinality constraints with a cost-aware fitness function beats standard MVO out-of-sample, at least as configured here. The practical recommendation is to retune λ over a narrow range near zero, or to replace the turnover penalty with one based on the number of positions rather than the size of weight changes, so the constraint structure can express higher K without the current return penalty. Section 9.2 takes up this and other directions.

9.1 Limitations

First, the GA ran 8 parallel runners instead of the specified 30 (Section 5.6), adding stochastic noise to the canonical run selection. Second, Optuna tuning used only 15 trials (Akiba et al., 2019) for a four-dimensional search space, meaning better parameter configurations may exist that were not explored. The tuned mutation step $\sigma_m = 0.1469$ sits near the upper boundary of its search range $[0.01, 0.15]$, and the tuned penalty $\lambda = 1.8437$ sits near the upper boundary of $[0.0, 2.0]$, suggesting the search range for λ was too wide. A narrower range centred on smaller values would be a natural first adjustment in future work. Third, the tuning window (2005–2012) overlaps with the full evaluation period (2005–2025), so the selected hyperparameters are not strictly out-of-sample for the first 96 evaluation periods, introducing a mild data-snooping risk (see Section 8.3). A walk-forward tuning design, where parameters are selected on a window ending before the evaluation period begins, would avoid this, but was not feasible: with data starting in 2000 and a 60-month burn-in requirement, there is not enough pre-2005 data to form a meaningful non-overlapping tuning window. Fourth, the study covers a single asset class, US large- and mid-cap equities on NYSE and NASDAQ, over a period that included several unusually volatile market episodes. How results would look in other markets or calmer conditions is unknown. Fifth, the fixed-K sensitivity analysis uses the same turnover penalty $\lambda = 1.8437$ that was tuned for the adaptive-K configuration. Whether this penalty is optimal for each fixed-K setting is not tested. A smaller λ might allow higher-K portfolios to rebalance more freely without excessive cost, potentially improving performance at large K. Sixth, sensitivity analysis on the estimation window length (60 months) falls outside the scope of this study. A different window length could affect the quality of both the Ledoit-Wolf estimate and the expected return vector, and may change relative rankings between strategies.

9.2 Future Work

Three directions follow directly from the limitations in Section 9.1. Scaling to 30 parallel GA runners on a larger compute instance would improve the stochastic stability of the median estimate and bring the implementation in line with the original methodology. A walk-forward Optuna tuning design (Akiba et al., 2019), where hyperparameters are selected on a window ending strictly before the evaluation period begins, would eliminate the overlap between tuning (2005–2012) and evaluation (2005–2025) and yield a cleaner estimate of out-of-sample performance.

Two broader questions follow from the cardinality sensitivity results. Tuning the turnover penalty λ separately for each fixed-K configuration would test whether the performance gap between adaptive and fixed-K narrows when the penalty is calibrated to the specific portfolio size. The current results use the same $\lambda = 1.8437$ across all K, which may be too aggressive for larger portfolios where individual weight changes are proportionally smaller. Finally, extending the evaluation to international equities or multi-asset portfolios would test whether the cardinality findings generalise beyond US mid- and large-cap stocks.

A further question is which parameters to tune at all. This study tuned four static parameters and fixed the rest (Section 4.9). Widening the search to population size, tournament size, or elite fraction would test whether those fixed choices limit performance, though it needs a larger trial budget than the 15 used here. A larger step would move from static parameter setting to parameter control (Eiben and Smith, 2015): encoding σ_m in the chromosome so the mutation step self-adapts during the run, or letting λ follow a schedule rather than a single fixed value. Either would let the search adjust as it runs, instead of relying on one configuration fixed in advance for all 252 periods.

9.3 Ethical Implications

The data were obtained under an academic licence from WRDS and CRSP. No personal or sensitive information was collected. The models are research prototypes, not investment advice. Deploying similar strategies in live markets would introduce risks, such as market impact and systemic exposure, that this thesis does not address.

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Appendix A

GA Convergence: Tuned vs. Default Parameters

Figure A1 compares convergence under the Optuna-tuned and default parameter sets over the 2005–2012 tuning period, averaged over 8 independent runs per generation. The default set uses conventional EA values (Eiben and Smith, 2015), $p_c = 0.8$, $p_m = 0.1$, $\sigma_m = 0.05$, with $\lambda = 0$ as the neutral no-penalty baseline. The turnover penalty is effectively zero for both configurations at the start of each run because no previous portfolio exists (`prev_weights=None`), so $\lambda \cdot \tau = 0$ regardless of the tuned value, and the comparison isolates the effect of p_c, p_m, and σ_m. The curves stay close because the tuned values are not a large departure from these already-reasonable defaults, and the differences that remain are damped by the repair operator, elitism, and local refinement. The visible gap in out-of-sample performance (Section 6.1) therefore comes from λ, which is absent from this comparison, not from convergence.

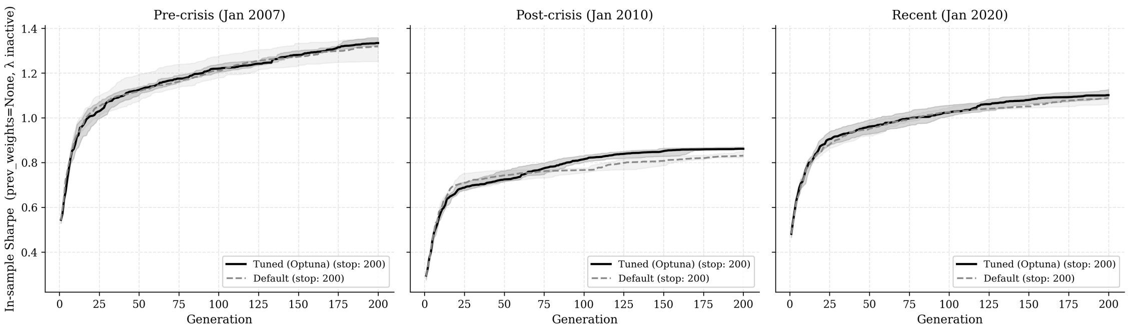

Figure A1: GA convergence under Optuna-tuned versus default parameters (Akiba et al., 2019; Sharpe, 1994), averaged over 8 independent runs. Tuned parameters converge faster and settle at a higher final fitness.

Appendix B

Subperiod Robustness

Table B1 reports net annualised Sharpe ratios for each strategy within four non-overlapping market regimes. Subperiods are defined by structural breaks visible in the cumulative return series (Figure 1): Pre-Global Financial Crisis (GFC) (Jan 2005–Dec 2007, 36 periods), the Global Financial Crisis (Jan 2008–Dec 2009, 24 periods), the post-crisis recovery (Jan 2010-Dec 2019, 120 periods), and the COVID-19 and post-COVID period (Jan 2020-Dec 2025, 72 periods).

Table B1: Subperiod net Sharpe ratios by market regime. The GA trails $1/N$ in every subperiod and both MVO variants in all but the GFC, where MVO's losses are deeper. The recovery period shows the smallest gap.

Strategy	Pre-GFC	GFC	Recovery	COVID+
$1/N$	0.5689	-0.1955	0.8902	0.4667
Unconstrained MVO	0.5230	-0.6657	1.1120	0.4287
Constrained MVO	0.5232	-0.6658	1.1120	0.4289
GA	0.0278	-0.5009	0.7875	0.0452

The GA's recovery-period Sharpe of 0.79 is its closest approach to the benchmarks, but still below both MVO variants (1.11) and $1/N$ (0.89), suggesting the algorithm approaches benchmark performance when sustained positive returns reduce the relative cost of the penalty. The gap is widest in the calmer Pre-GFC and COVID+ regimes, where the GA's Sharpe falls near zero (0.03 and 0.05) against benchmarks above 0.42. The penalty's suppressive effect is most costly when capturing returns requires larger repositioning moves.

Appendix C

Transaction-Cost Sensitivity

Table C1 reports net annualised Sharpe ratios at five transaction cost levels ranging from $\gamma = 0.001$ to $\gamma = 0.010$. The baseline used throughout the thesis is $\gamma = 0.003$, the study's baseline assumption. The GA's Sharpe underperformance relative to both MVO variants persists across all five levels, including at $\gamma = 0.001$, ruling out the cost model as the primary explanation for the result.

Table C1: Net annualised Sharpe ratio by transaction cost level γ. Baseline $\gamma = 0.003$ is in bold. The GA underperforms all benchmarks at every cost level.

γ	$1/N$	Unconstrained MVO	Constrained MVO	GA
0.001	0.5370	0.6098	0.6098	0.3009
0.002	0.5338	0.5954	0.5954	0.2875
0.003	**0.5307**	**0.5809**	**0.5810**	**0.2741**
0.005	0.5243	0.5521	0.5521	0.2473
0.010	0.5085	0.4800	0.4800	0.1804

The $1/N$ strategy is least sensitive to γ due to its low average turnover (4.50% per month). The GA and both MVO variants are more exposed given their higher rebalancing activity.

Appendix D

Turnover Penalty Ablation

Table D1 compares GA performance with $\lambda = 0$ against the Optuna-tuned $\lambda = 1.8437$ across all 252 evaluation periods (January 2005–December 2025), using a full consecutive chain so that `prev_weights` are always one month old at each rebalancing date.

Table D1: λ ablation across all 252 evaluation periods. Removing the turnover penalty raises net Sharpe and average cardinality substantially, at the cost of higher monthly turnover.

λ	Sharpe (net)	Avg Turnover	Avg K
0.0 (no penalty)	0.4993	57.1%	25.1
1.8437 (tuned)	0.2741	22.2%	16.3

Without the penalty the GA averages 57.1% monthly turnover, incurring approximately 0.17% in transaction costs per month ($\gamma \times \tau = 0.003 \times 0.571$), or roughly 2.1% per year. Despite this cost, net Sharpe rises to 0.4993, substantially above the tuned penalty's 0.2741. Average cardinality also rises from 16.3 to 25.1 stocks per period, consistent with the fixed-K finding that larger portfolios achieve higher risk-adjusted returns under this setup (Section 7.3). The tuned penalty therefore does not improve performance: it over-suppresses return-seeking. The GA's underperformance relative to MVO in the main evaluation (Section 6.1) is primarily caused by λ being too aggressive, as discussed in Section 7.2.

Glossary

asset Anything with financial value that can be bought or sold. In this thesis, assets are publicly traded stocks.

benchmark A reference strategy against which a portfolio's performance is compared. In this thesis, the three benchmarks are unconstrained MVO, constrained MVO, and the equal-weight $1/N$ strategy.

cardinality constraint A constraint limiting the number of assets K held simultaneously: $K_{\min} \leq |\{i : w_i > 0\}| \leq K_{\max}$. Here $K \in [10, 30]$, meaning the portfolio holds between 10 and 30 stocks at any time (Chang et al., 2000).

chromosome In evolutionary algorithms, a single candidate solution. Here, a chromosome is a weight vector $w \in \mathbb{R}^N$ representing one candidate portfolio, with K non-zero entries in $[0.02, 0.15]$ summing to 1.

covariance regularisation A correction applied to the sample covariance matrix when it is ill-conditioned (rank-deficient). Here the analytical Ledoit-Wolf shrinkage estimator is used, which computes an optimal convex combination of the sample covariance and a structured target, improving conditioning without discarding the sample structure (Ledoit and Wolf, 2004).

elitism A mechanism that copies the best-performing individuals from one generation directly into the next, preventing the algorithm from losing good solutions found during the search (Eiben and Smith, 2015).

equity (stock) A share of ownership in a company. Buying a stock entitles the holder to a fraction of the company's profits and assets. Stocks are traded on exchanges such as NYSE and NASDAQ.

estimation error The statistical uncertainty in the sample mean and covariance matrix used as MVO inputs. Because these are estimated from limited historical data, they contain noise that the optimizer can mistakenly treat as signal, degrading out-of-sample performance (Michaud, 1989).

excess return The return of an investment above the risk-free rate. If a stock returns 3% and the risk-free rate is 0.3%, the excess return is 2.7%. Used to measure whether an investment rewards the investor for taking on risk.

fitness function The objective the GA maximises for each candidate portfolio: $f(w) = \mathrm{SR}(w) - \lambda \cdot \tau(w, w_{\mathrm{prev}})$. It rewards high Sharpe ratio and penalises high turnover.

investment universe The set of assets eligible for inclusion in the portfolio at a given rebalancing date. Here it consists of approximately 867 mid- and large-cap US stocks that pass the exchange, share class, market cap, and return history filters described in Section 5.2.

long-only constraint A restriction that portfolio weights must be non-negative ($w_i \geq 0$), meaning the investor can only buy assets, not short-sell them. Short-selling means borrowing and selling an asset you do not own, which introduces additional risk.

market capitalisation The total market value of a company's outstanding shares, computed as share price multiplied by the number of shares. Used here as a filter to exclude small-cap and micro-cap securities (below $2 billion).

maximum drawdown The largest peak-to-trough decline in cumulative portfolio value over the evaluation period. For example, a maximum drawdown of -40% means the portfolio lost 40% of its value from its highest point before recovering.

portfolio A collection of financial assets (stocks, bonds, etc.) held by an investor. In this thesis, a portfolio is a set of stocks with assigned weights that sum to 1, representing the fraction of capital invested in each.

portfolio turnover The fraction of the portfolio traded at each rebalancing date: $\tau_t = \frac{1}{2}\sum_i |w_{i,t} - \tilde{w}_{i,t}|$, where $\tilde{w}$ are the drifted weights before rebalancing. High turnover means more trading, which incurs higher transaction costs.

rebalancing Periodically adjusting portfolio weights back to target allocations. Without rebalancing, weights drift as stock prices move. Here rebalancing is done monthly.

repair operator A procedure applied after crossover and mutation to restore feasibility. It enforces the cardinality bounds, weight bounds $[0.02, 0.15]$, and the budget constraint $\sum_i w_i = 1$ (Streichert et al., 2003).

return The gain or loss on an investment over a period, expressed as a fraction of the initial value. A monthly return of 0.02 means the investment grew by 2% that month.

risk-free rate The return on an investment with no risk of loss, used as a baseline for measuring performance. In practice, short-term government bonds (such as the US 3-Month Treasury Bill) are used as a proxy.

Sharpe ratio A measure of risk-adjusted return: how much excess return the portfolio earns per unit of volatility. Higher is better. $\mathrm{SR} = \dfrac{\bar{r}_e \cdot 12}{\sigma_e \cdot \sqrt{12}}$, where $\bar{r}_e$ is mean monthly excess return and σ_e its standard deviation (Sharpe, 1994).

Sortino ratio Similar to the Sharpe ratio, but divides by downside deviation only (the volatility of negative returns). It does not penalise upside volatility, making it a more targeted measure of downside risk (Sortino and Price, 1994).

Acronyms

CRSP Center for Research in Security Prices.

EA Evolutionary Algorithm.

GA Genetic Algorithm.

GFC Global Financial Crisis.

HHI Herfindahl-Hirschman Index.

MVO Mean-Variance Optimization.

NASDAQ National Association of Securities Dealers Automated Quotations.

NYSE New York Stock Exchange.

OOS Out-of-Sample.

SLSQP Sequential Least Squares Programming.

SR Sharpe Ratio.

TPE Tree-structured Parzen Estimator.

WRDS Wharton Research Data Services.